

Joseph Walton

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Professional Summary

Experienced machine learning engineer with over 8 years of expertise in machine learning, AI engineering & software engineering. Demonstrable success in building and deploying advanced AI models and production-grade ML systems with significant business value. Passionate about developing secure, responsible AI systems. Possesses excellent stakeholder management skills & ability to guide organisations through AI transformation.

Technical Skills

- **Programming Languages:** Python, SQL, Java
 - **LLM ecosystem:** AzureOpenAI, Huggingface, Langchain, Langgraph
 - **Machine Learning Frameworks:** TensorFlow, Pytorch, Scikit-learn
 - **Data Analysis:** Apache Spark, Pandas, Polars, NumPy, Matplotlib, Seaborn
 - **Tools:** Git, Docker, Jupyter, MLFlow
 - **Cloud Platforms:** Azure, AWS, Google Cloud Platform
 - **Databases:** MySQL, PostgreSQL, Elasticsearch
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Professional Experience

IBM

Lead Data Scientist

Contract role

Contract role with central government.

- Rescuing failing AI projects through implementing best practices. (Evaluations, Observability)
 - Designed and delivered an LLM powered data processing assistant saving 2 hours per case worker per day for a saving of ~£2M a year.
 - Wrote a Claude Skill to optimise Human-in-the-Loop data ingestion for writing Evals for LLM products, time saved has been ~1 hour per case in the Eval suite.
 - Rewrote a data matching algorithm in Polars from Apache Spark. This reduced cloud costs by ~£10k per month, reducing run time from 2 hours to 10 minutes.
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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - Present

Developing NLP solutions with a focus on improving outcomes of trade mark applications.

- Developed a product roadmap for AI applications in the IP space

- Collaboration with multiple international IP offices communicating how the UK IPO is implementing AI in the IP space, to both technical & non-technical stakeholders.
- Member of cross-government AI working group, contributing to the development of AI policy and best practices.
- Developed Generative AI powered applications within a service-oriented architecture using Langchain, AzureOpenAI & Azure Functions.
- Built a RAG agent to help novice users identify relevant Goods & Service terms.
- Conducted red teaming exercises to evaluate and improve the robustness of Generative AI applications against adversarial attacks.
- Implemented vector search using an Elasticsearch backend, tuned HNSW implementation to balance throughput & recall.
- Data driven refinement of our search & recommendation systems, improving relevance metrics by 20%.
- Implemented an end to end MLOps strategy to improve model deployment times to <1 day across multiple environments.
- Architected & implemented ETL pipelines using Polars, replacing an expensive Azure Databricks implementation, reducing compute costs by 50% & enabling zero downtime deployments.

Tech Stack: Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led a team focused on transforming business operations through secure and innovative ML solutions.

- Led the design and implementation of ML pipelines using MLOps best practices, enhancing system integrity and efficiency. This improved time to market for new models to <1 day, allowing for client specific model deployments.
- Developed and deployed NLP and computer vision models for intellectual property protection, including a multilingual BERT-based model for detecting IP infringements across 90+ languages.
- Implemented a recommendation engine based on text and image similarity, enhancing the detection of IP violations.
- Mentored a software engineer to help her transition into a machine learning role.

Tech Stack: Vertex AI Pipelines, Tensorflow, Pyspark, Google Cloud Platform, MLFlow

Machine Learning Engineer

November 2018 - February 2020

Focused on the research and development of ML models for IP protection.

- Engineered a production system for predicting IP infringement, leveraging NLP techniques and deploying it within a secure, containerized environment. Increased number of enforced infringements by 10% per analyst (the company employed ~200 analysts) .
- Built & deployed an image similarity algorithm to identify IP infringing items. This feature was instrumental in securing a major contract with a blue-chip client, contributing to 10% of Annual Recurring Revenue (ARR).

Tech Stack: TensorFlow, FastText, Google Cloud Platform, MLFlow

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

Contributed to national-scale data integration projects with a focus on secure data handling.

- Developed ML models and data pipelines to link and process government administrative data, supporting national survey efforts.
- Worked on the UK Census 2021 project, developing a highly scalable backend for the Census data collection system.

Tech stack : Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018

Seconded to Barclays for a project focused on using payment data to create economic indicators.

- Worked on a high-impact project involving secure and ethical use of financial data to develop regional economic models.

This was a continuation of my work as part of the winning team of an ONS X Barclays Hackathon.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

Engaged in full-stack development of big data processing systems for multiple clients in the government and defense sectors.

Tech stack: Java, Python, PySpark, AngularJs

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & awards

Winner of ONS x Barclays Data Hackathon

Using payments data to create more granular and timely estimates for GVA, (Gross Value Added).